

Dokumentasi Alur Pengerjaan Project

1. Download Data Set

The screenshot shows the Kaggle dataset page for 'Brazilian E-Commerce Public Dataset by Olist'. The page includes a sidebar with navigation links, a search bar, and a main content area. The dataset is described as having 100,000 orders with product, customer, and review information. It is a public dataset by Olist, updated 5 years ago, with 4180 views and 756 code submissions. The 'About Dataset' section provides a welcome message and details about the data's origin and anonymization. The 'Usability' section shows a score of 10.00. The 'License' is CC BY-NC-SA 4.0. The 'Expected update frequency' is 'Never'. The 'Tags' section includes 'Business'.

Brazilian E-Commerce Public Dataset by Olist
100,000 Orders with product, customer and reviews info

About Dataset

Brazilian E-Commerce Public Dataset by Olist

Welcome! This is a Brazilian ecommerce public dataset of orders made at Olist Store. The dataset has information of 100k orders from 2016 to 2018 made at multiple marketplaces in Brazil. Its features allows viewing an order from multiple dimensions: from order status, price, payment and freight performance to customer location, product attributes and finally reviews written by customers. We also released a geolocation dataset that relates Brazilian zip codes to lat/long coordinates.

This is real commercial data, it has been anonymised, and references to the companies and partners in the review text have been replaced with the names of Game of Thrones great houses.

Usability
10.00

License
CC BY-NC-SA 4.0

Expected update frequency
Never

Tags
Business

2. Extract File

The screenshot shows a Windows File Explorer window with the 'Downloads' folder selected. The folder contains several CSV files, all dated 21/05/2026 17:31. The files are listed in a table with columns for Name, Date modified, Type, and Size.

Name	Date modified	Type	Size
olist_orders_dataset.csv	21/05/2026 17:31	Microsoft Excel Co...	17.242 KB
olist_products_dataset.csv	21/05/2026 17:31	Microsoft Excel Co...	2.324 KB
olist_sellers_dataset.csv	21/05/2026 17:31	Microsoft Excel Co...	171 KB
product_category_name_translation.csv	21/05/2026 17:31	Microsoft Excel Co...	3 KB
olist_geolocation_dataset.csv	21/05/2026 17:31	Microsoft Excel Co...	59.838 KB
olist_order_items_dataset.csv	21/05/2026 17:31	Microsoft Excel Co...	15.077 KB
olist_order_payments_dataset.csv	21/05/2026 17:31	Microsoft Excel Co...	5.642 KB
olist_order_reviews_dataset.csv	21/05/2026 17:31	Microsoft Excel Co...	14.113 KB
olist_customers_dataset.csv	21/05/2026 17:31	Microsoft Excel Co...	8.823 KB

3. Penggabungan Table menggunakan python dan pandas

```
01_RFM_Analysis.ipynb •
01_RFM_Analysis.ipynb > import pandas as pd
Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline ...

import pandas as pd
import datetime as dt

# Membaca 3 file dataset utama (karena filenya ada di luar, kita langsung panggil namanya).
df_orders = pd.read_csv('olist_orders_dataset.csv')
df_payments = pd.read_csv('olist_order_payments_dataset.csv')
df_customers = pd.read_csv('olist_customers_dataset.csv')

# Filter: Hanya mengambil kolom yang kita butuhkan agar prosesnya cepat.
df_orders = df_orders[['order_id', 'customer_id', 'order_purchase_timestamp']]
df_payments = df_payments[['order_id', 'payment_value']]
df_customers = df_customers[['customer_id', 'customer_unique_id']]

# Menggabungkan (Merge) ketiga tabel menggunakan fungsi join/merge.
# Gabung orders dan customers
df_merge_1 = pd.merge(df_orders, df_customers, on='customer_id')
# Gabung hasilnya dengan payments.
df_final = pd.merge(df_merge_1, df_payments, on='order_id')

# Mengubah tipe data tanggal (dari teks menjadi format Datetime yang bisa dihitung).
df_final['order_purchase_timestamp'] = pd.to_datetime(df_final['order_purchase_timestamp'])

# Tampilkan hasil gabungan.
print(f"Total baris data gabungan: {df_final.shape[0]}")
display(df_final.head())

[1] ✓ 10.6s

Total baris data gabungan: 103886

order_id customer_id order_purchase_timestamp customer_unique_id payment_value
0 e481f51cbd54678b7cc49136f2d6af7 9ef432eb6251297304e76186b10a928d 2017-10-02 10:56:33 7c396fd4830fd04220f754e42b4e5bff 18.12
1 e481f51cbd54678b7cc49136f2d6af7 9ef432eb6251297304e76186b10a928d 2017-10-02 10:56:33 7c396fd4830fd04220f754e42b4e5bff 2.00
2 e481f51cbd54678b7cc49136f2d6af7 9ef432eb6251297304e76186b10a928d 2017-10-02 10:56:33 7c396fd4830fd04220f754e42b4e5bff 18.59
3 53cdb2f8bc7dce0b6741e2150273451 b0830fb4747a6cd20dea0b8c802d7ef 2018-07-24 20:41:37 af07308b275d755c9edb36a90c618231 141.46
4 47770eb9100c2d0c44946d9cf07ec65d 41ce2a54c0b03bf3443c3d931a367089 2018-08-08 08:38:49 3a653a41f6f9f3d2a113cf8398680e8 179.12
```

4. Menghitung nilai RFM (Recency, Frequency, Monetary) untuk setiap pelanggan.

```
01_RFM_Analysis.ipynb •
01_RFM_Analysis.ipynb > # Menentukan tanggal acuan (snapshot date).
Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline ...

# Menentukan tanggal acuan (snapshot date).
# Kita ambil 1 hari setelah transaksi terakhir di dataset sebagai "hari ini".
snapshot_date = df_final['order_purchase_timestamp'].max() + dt.timedelta(days=1)

# Menghitung RFM untuk setiap pelanggan (berdasarkan KTP/customer_unique_id).
df_rfm = df_final.groupby('customer_unique_id').agg({
    'order_purchase_timestamp': lambda x: (snapshot_date - x.max()).days, # Recency: Selisih hari dari transaksi terakhir.
    'order_id': 'nunique', # Frequency: Jumlah order yang unik.
    'payment_value': 'sum' # Monetary: Total uang yang dihabiskan.
}).reset_index()

# Mengubah nama kolom agar mudah dibaca di Power BI nanti.
df_rfm.rename(columns={
    'order_purchase_timestamp': 'Recency',
    'order_id': 'Frequency',
    'payment_value': 'Monetary'
}, inplace=True)

# Tampilkan hasilnya.
print(f"Total pelanggan unik yang dianalisis: {df_rfm.shape[0]}")
display(df_rfm.head())

[2] ✓ 32.2s

Total pelanggan unik yang dianalisis: 96095

customer_unique_id Recency Frequency Monetary
0 0000366fb3b9a7992b78c76cdf3221e2 161 1 141.90
1 0000b849f77a49e4a4ce2b2a4ca5be3f 164 1 27.19
2 0000f46a3911fa3c080544483337064 586 1 86.22
3 0000f6ccb0745a6a4b88665a16c9f078 370 1 43.62
4 0004aac84e0df4da2b147fca70c78255 337 1 196.89
```

5. Export Data ke CSV

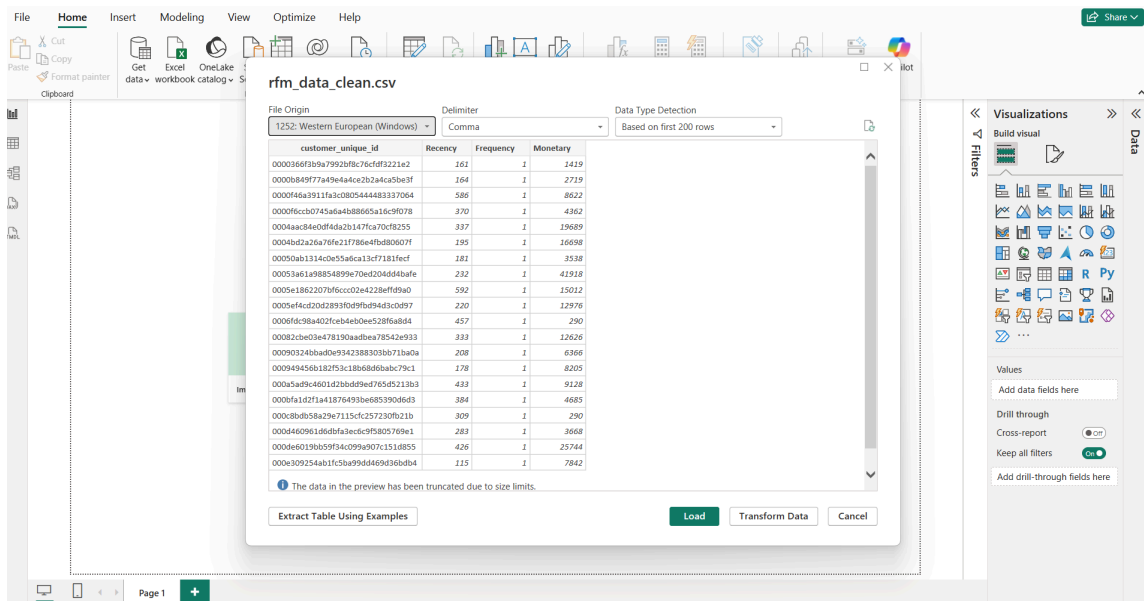
```
# Menyimpan tabel RFM yang sudah bersih ke dalam file CSV baru.
df_rfm.to_csv('rfm_data_clean.csv', index=False)

print("Data berhasil disimpan sebagai 'rfm_data_clean.csv'!")
```

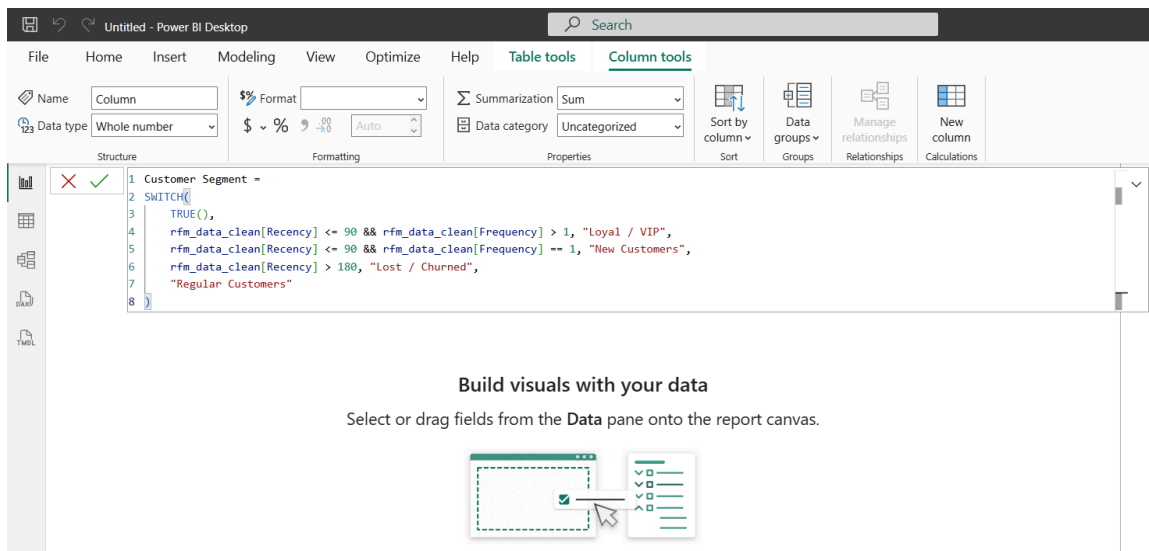
[3] ✓ 0.2s

... Data berhasil disimpan sebagai 'rfm_data_clean.csv'!

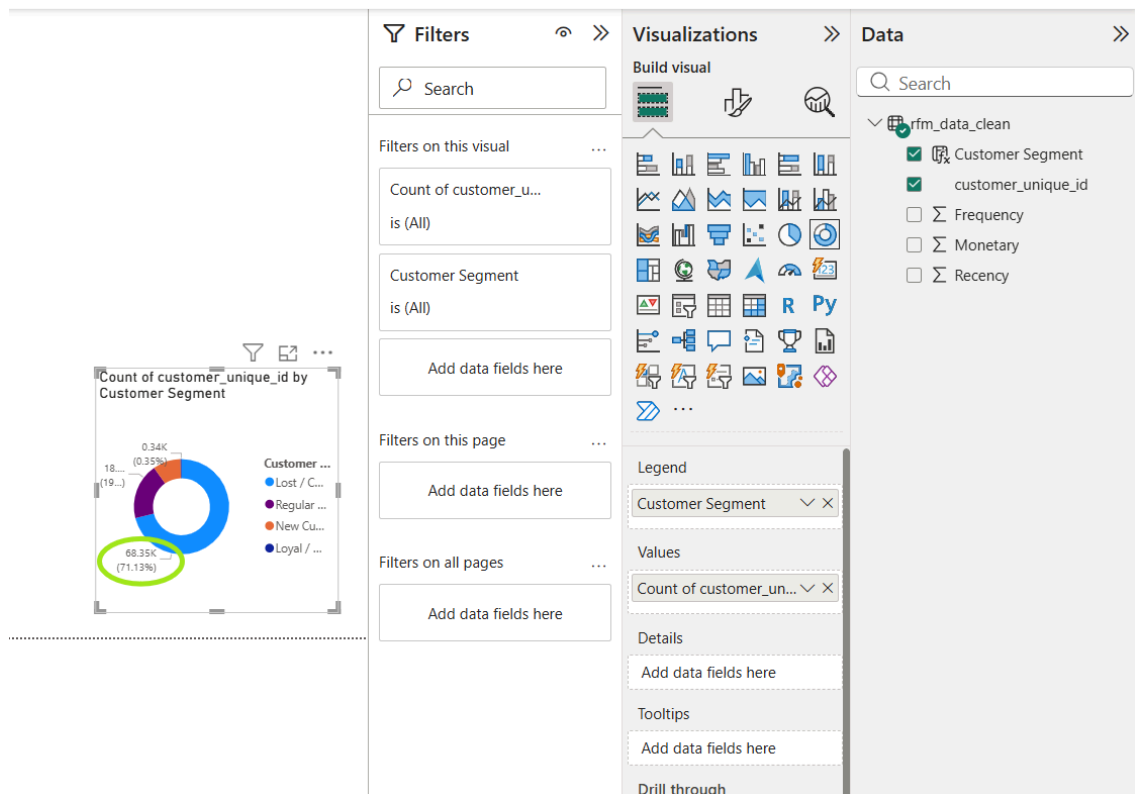
6. Tahap Visualisasi Menggunakan power BI (Import Data)



7. Membuat Kolom Segmen (DAX)



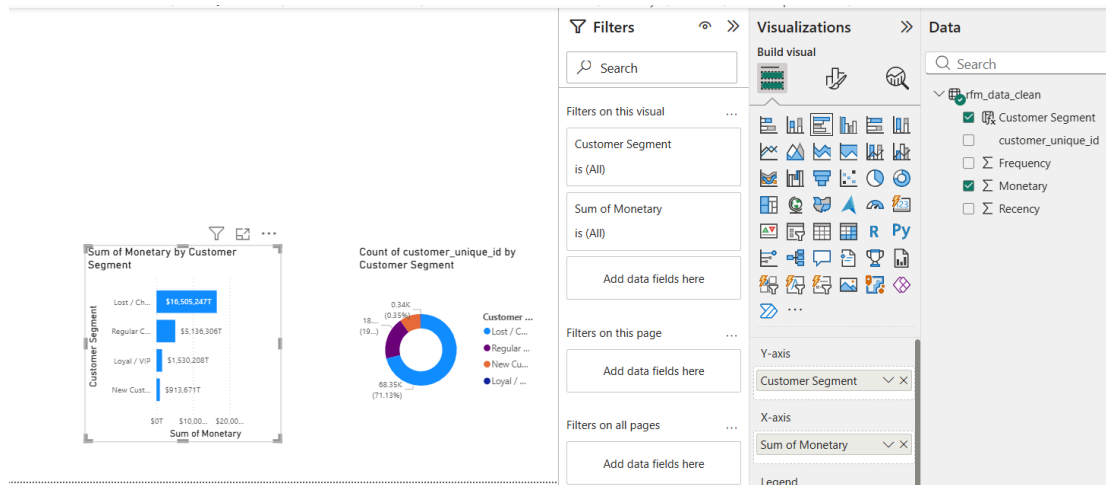
8. Membuat Visualisasi Pertama (Donut Chart)



Key Insight: Tingkat Churn Sangat Tinggi (71%)

- **Temuan Data:** Sebanyak 71% pelanggan berada di segmen **Lost / Churned** (tidak bertransaksi selama lebih dari 6 bulan). Ini menunjukkan bahwa perusahaan unggul dalam akuisisi pengguna baru, namun memiliki kendala besar dalam retensi (*customer retention*).
- **Rekomendasi Bisnis:** Tim Marketing perlu segera mengeksekusi kampanye reaktivasi (*Reactivation Campaign*). Contohnya: Mengirimkan *email marketing* berisi diskon "Welcome Back" secara eksklusif kepada segmen ini untuk memulihkan pendapatan yang hilang.

9. Memunculkan Angka pada Grafik



Dari analisis RFM, kita menemukan sebuah ancaman sekaligus peluang besar. Segmen **Lost / Churned** mendominasi basis pelanggan sebesar **71%**. Namun yang mengejutkan, mereka juga merupakan kelompok dengan historis pengeluaran (Monetary) tertinggi, yaitu mencapai **\$16.5 Juta**. Artinya, perusahaan sedang duduk di atas "harta karun" pelanggan lama yang dibiarkan tidur. Rekomendasi utama saya adalah merealokasi budget marketing dari sekadar mencari pelanggan baru, menjadi kampanye retargeting agresif untuk menghidupkan kembali pelanggan lama ini.